

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Number	: CO3_AT2
Assessment Tool Weightage	: 35%
Assessment Tool	: Design Task
Duration	: 180 Minutes
Marks	: 100 Marks
Type of Assessment Conduction	: At Home
Evaluation Type	: GitHub
Result Generation	: Students design and justify appropriate wireless communication solutions by applying engineering principles to meet specified requirements.
Topic Covered	: Spectrum management, licensed/unlicensed spectrum, dynamic spectrum access, cognitive radio, OFDMA, resource block allocation, GFDMA, NOMA, power-domain multiplexing, SIC, and IoT applications. Advanced waveforms and wireless technologies including GFDM, FBMC, UFMC, mmWave propagation, path loss and penetration, millimeter-wave antenna systems, MIMO, and Massive MIMO fundamentals.

Conduction of Assessment Procedure:

- In the Design Based Assessment, each student will be given one comprehensive design question carrying 100 marks, making the total assessment 100 marks.
- The evaluation will focus on the student's ability to understand the problem, develop an appropriate design solution, apply relevant engineering concepts, and present a clear implementation approach.
- Marks will also be awarded for innovation, accuracy, technical justification, and the ability to defend the design during discussion or viva voce, ensuring a complete assessment of practical and analytical design skills.
- Report will be uploaded in gitHub and marks will be awarded.

19.Design a spectrum sharing model between LTE and 5G users. (BL6)

Design Task

Design a Spectrum Sharing Model Between LTE and 5G Users

Aim

To design and simulate a spectrum sharing model between LTE and 5G users and compare the bandwidth allocation.

Introduction

Spectrum sharing is the process of allowing multiple wireless communication systems to use the same radio frequency spectrum efficiently. Since the radio spectrum is limited, LTE (4G) and 5G networks can share the available spectrum to improve network performance and support more users. This helps in maximizing spectrum utilization and reducing interference.

Theory

LTE and 5G are mobile communication technologies that require radio spectrum for transmitting and receiving data. During the transition from LTE to 5G, operators often use Spectrum Sharing so that both networks can operate simultaneously within the same frequency band.

In this model:

- LTE users require a certain amount of bandwidth.
- 5G users require higher bandwidth due to higher data rates.
- The total available spectrum is shared between LTE and 5G users.
- If the total spectrum demand exceeds the available spectrum, bandwidth is allocated proportionally to ensure fair usage.

This approach improves spectrum efficiency and provides better Quality of Service (QoS) for users.

Objectives

- To understand the concept of spectrum sharing.
- To allocate spectrum between LTE and 5G users.
- To simulate spectrum allocation using MATLAB.
- To analyse spectrum utilization and bandwidth distribution

Algorithm

1. Define the total available spectrum.
2. Specify the number of LTE and 5G users.
3. Assign bandwidth requirements for LTE and 5G users.
4. Calculate the total spectrum required.
5. Compare the required spectrum with the available spectrum.
6. Allocate spectrum equally or proportionally based on availability.
7. Display the allocated spectrum.
8. Plot the spectrum allocation using a bar graph.

CODE

```
clc;
clear;
close all;

% Total Available Spectrum (MHz)
total_spectrum = 100;

% Number of Users
lte_users = 20;
g5_users = 30;

% Bandwidth Requirement per User (MHz)
lte_bw = 2;
g5_bw = 3;

% Required Spectrum
lte_required = lte_users * lte_bw;
g5_required = g5_users * g5_bw;

fprintf('Total Spectrum Available = %d MHz\n', total_spectrum);
fprintf('LTE Required Spectrum = %d MHz\n', lte_required);
fprintf('5G Required Spectrum = %d MHz\n', g5_required);

% Spectrum Sharing
if (lte_required + g5_required) <= total_spectrum

    allocated_lte = lte_required;
    allocated_5g = g5_required;

else

    total_required = lte_required + g5_required;

    allocated_lte = (lte_required/total_required)*total_spectrum;
    allocated_5g = (g5_required/total_required)*total_spectrum;

end

fprintf('\nAllocated LTE Spectrum = %.2f MHz\n', allocated_lte);
fprintf('Allocated 5G Spectrum = %.2f MHz\n', allocated_5g);

% Utilization
utilization = ((allocated_lte+allocated_5g)/total_spectrum)*100;

fprintf('Spectrum Utilization = %.2f %%\n', utilization);

% Plot
figure;

bar([allocated_lte allocated_5g])

grid on
```

```
set(gca,'XTickLabel',{'LTE','5G'})  
  
ylabel('Allocated Spectrum (MHz)')  
  
xlabel('Network')  
  
title('Spectrum Sharing between LTE and 5G')
```

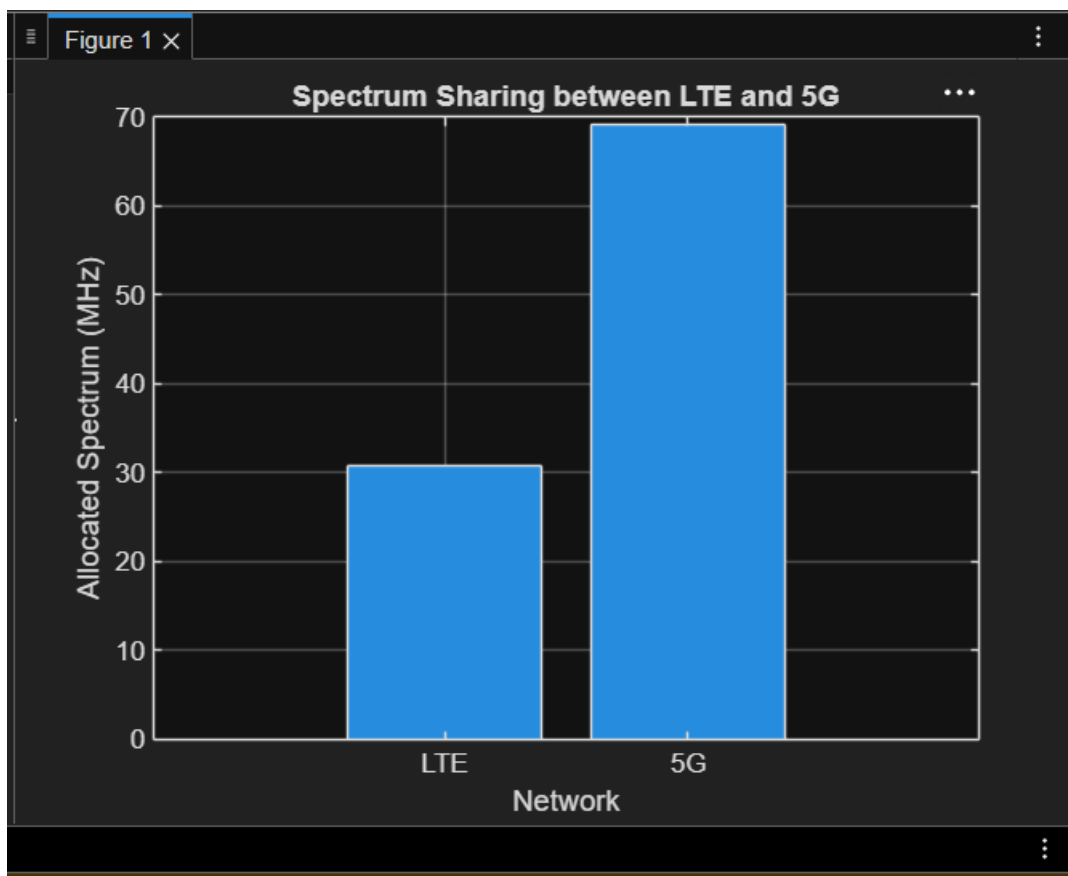
Output Graph

The graph displays:

- X-axis → LTE and 5G
- Y-axis → Allocated Spectrum (MHz)

Example:

Allocated Spectrum



New to MATLAB? See resources for [Getting Started](#).

```
Total Spectrum Available = 100 MHz
LTE Required Spectrum = 40 MHz
5G Required Spectrum = 90 MHz

Allocated LTE Spectrum = 30.77 MHz
Allocated 5G Spectrum = 69.23 MHz
Spectrum Utilization = 100.00 %
>>
```

Inference

- LTE users require 40 MHz, while 5G users require 90 MHz.
- Since the total requested spectrum (130 MHz) exceeds the available spectrum (100 MHz), the system performs proportional spectrum sharing.
- LTE is allocated approximately 30.77 MHz, and 5G is allocated approximately 69.23 MHz.
- The available spectrum is fully utilized (100% utilization) while ensuring fair allocation between LTE and 5G users.

Result

The MATLAB simulation successfully demonstrates spectrum sharing between LTE and 5G users. The available spectrum is allocated based on user demand, and the results are displayed graphically, showing efficient utilization of the available bandwidth.

Conclusion

The spectrum sharing model allows LTE and 5G users to efficiently share the available frequency spectrum. The MATLAB simulation shows that proportional spectrum allocation improves utilization while supporting both networks. This technique is widely used in modern wireless communication systems to ensure efficient and fair spectrum management.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	25%	Demonstrates deep understanding of design objectives, constraints, and context (25)	Shows good understanding with minor gaps (15)	Partial understanding; misses key aspects (10)	Lacks understanding of the design context (5)
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps (25)	Identifies key issues with reasonable analysis (15)	Limited critique; mostly descriptive feedback (10)	No meaningful critique provided (5)
Justification	25%	Supports critique with strong reasoning, evidence, and relevant principles (25)	Provides justification with some supporting evidence (15)	Weak or unclear justification (10)	No justification for feedback (5)
Suggestions for Improvement	25%	Offers innovative, practical, and well-justified improvement suggestions (25)	Provides useful suggestions with minor gaps (15)	Suggestions are limited or partially relevant (10)	No constructive suggestions provided (5)